

The aqueous extract of Hibiscus sabdariffa calices modulates the production of monocyte chemoattractant protein-1 in humans.

.

Diet supplementation and/or modulation is an important strategy to significantly improve human health. The search of plants as additional sources of bioactive phenolic compounds is relevant in this context. The aqueous extract of Hibiscus sabdariffa is rich in anthocyanins and other phenolic compounds including hydroxycitric and chlorogenic acids. Using this extract we have shown an effective protection of cultured peripheral blood mononuclear cells from the cellular death induced by H₂O₂ and a significant role in the production of inflammatory cytokines. In vitro, the extract promotes the production of IL-6 and IL-8 and decreases the concentration of MCP-1 in supernatants in a dose-dependent manner. In humans, the ingestion of an acute dose of the extract (10g) was well tolerated and decreased plasma MCP-1 concentrations significantly without further effects on other cytokines. This effect was not due to a concomitant increase in the antioxidant capacity of plasma. Instead, its mechanisms probably involve a direct inhibition of inflammatory and/or metabolic pathways responsible for MCP-1 production, and may be relevant in inflammatory and chronic conditions in which the role of MCP-1 is well established. If beneficial effects are confirmed in patients, Hibiscus sabdariffa could be considered a valuable traditional herbal medicine for the treatment of chronic inflammatory diseases with the advantage of being devoid of caloric value or potential alcohol toxicity. Copyright 2009 Elsevier GmbH. All rights reserved.